The ΔC_0 map for SATA: current map vs. look-up table — the essentials

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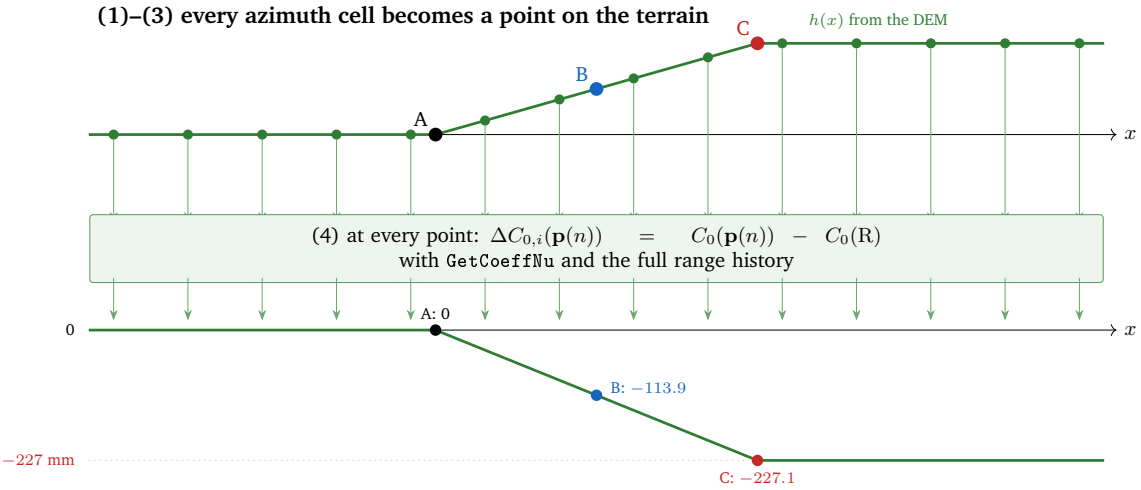
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Common to both. The SATA kernel is the same: sub-apertures of T_{sub} samples, triangular window, FFT; every bin gets a Doppler label $\hat{f}$ (whole band: $\hat{f} = \phi \in [-\mathrm{PRF}_{ch}/2, \mathrm{PRF}_{ch}/2)$; sub-band k: $\hat{f} = \phi + m\,\mathrm{PRF}_{ch}$ inside the window of band k), is mapped to an azimuth cell through the ruler

$$x = c + a(\hat{f}), \qquad a(f) = \frac{r_0}{\Delta x}\tan\Big[\arcsin\frac{\lambda f}{2v}\Big] \approx \frac{r_0 \lambda\,\mathrm{PRF}_{ch}}{2v^2}\,f,$$

and is multiplied by $e^{+j2\pi M[x]/\lambda}$. The number $\Delta C_{0,i}(\mathbf{p}) = C_0(\mathbf{p}) - C_0(\mathrm{R})$ (full range history, `GetCoeffNu`) is also the same in both. **The two methods differ only in what is written into the table $M[x]$.**

1) Look-up table



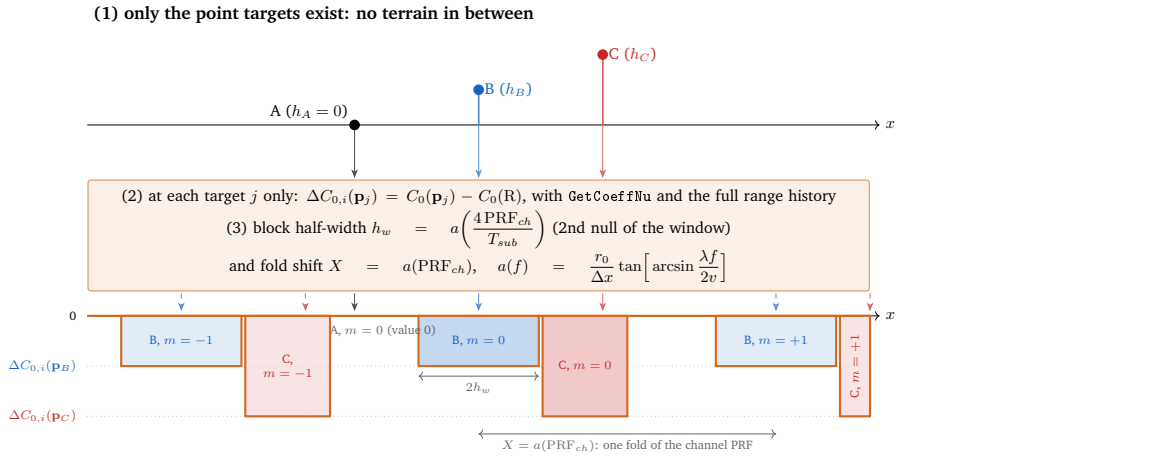
(5) $M_{\mathrm{LUT}}[n]$: one value per cell, the same table for every kernel, no copies

It needs the height over the whole scene. First $h(x)$ is taken from the DEM at every azimuth cell; then $\Delta C_0(x)$ is computed at every cell with the full range history, giving a table with one value per terrain position. At run time the kernel does $f \to \beta \to x$ for every bin and applies the table value at cell x. Because x comes from the label $\hat{f}$ the kernel assigns to the bin, not from the true Doppler, the table is read at the position of the *guess*: if the guess is off by one fold of PRF_{ch}, the kernel reads the terrain at $X = a(\mathrm{PRF}_{ch})$ from the target.

pipeline

points $\to$ $h(x)$ (DEM) $\to$ $\Delta C_0(x)$ at every cell $\to$ $\boxed{f \to \beta \to x}$ $\to$ $e^{+j2\pi\Delta C_0(x)/\lambda}$ (one table, the same for every kernel; no copies)

2) Current map: aliasing considered

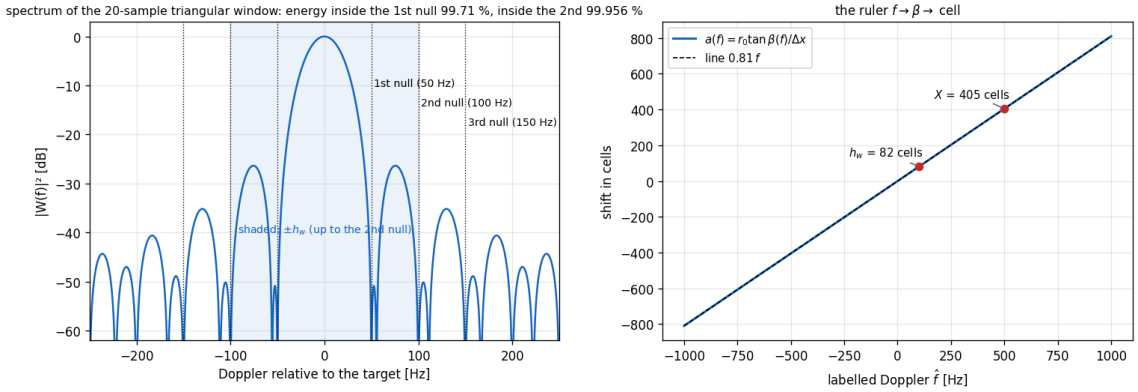


(4) $M_{\text{current}}[n]$: the value of target j in a block $|n - n_j - mX| \le h_w$ for every $m \in \mathcal{M}$; zero elsewhere

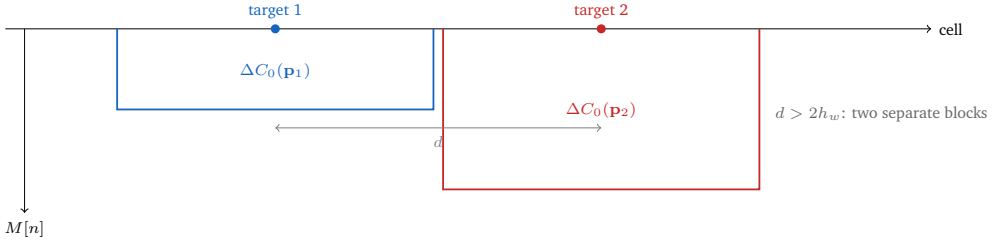
$\mathcal{M} = \left\{\mathrm{round}\frac{f_c - B_a/2}{\mathrm{PRF}_{ch}}, \ldots, \mathrm{round}\frac{f_c + B_a/2}{\mathrm{PRF}_{ch}}\right\}$, f_c = centre of the kernel's label window (0 for whole band, f_k for sub-band k), B_a = processed bandwidth;

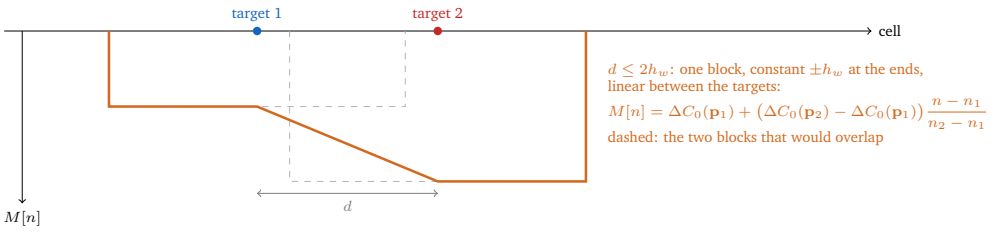
where two images overlap, the cell goes to the image whose core is nearer (here: C's block is cut by the $m = +1$ copy of A). This scene: $h_w = 82$ cells, $X = 405$ cells, $\mathcal{M}_{\text{whole}} = \{-1, 0, +1\}$.

It takes the channel aliasing into account: the energy of a target can be read at the target's position or at the fold positions, multiples mX away, depending on the label the kernel uses. So it writes the target's number at every position where the kernel may look for it: at the target's cell n_j and at the copies $n_j + mX$, with m in the set $\mathcal{M}$ allowed by the kernel's labels ($\{-1, 0, +1\}$ for whole band; a shifted set of four values for each sub-band). It also takes into account that the target's energy in the sub-aperture spectrum fills the main lobe and the first sidelobe of the window, up to the second null at $4\,\mathrm{PRF}_{ch}/T_{sub}$, which on the ruler is a half-width $h_w = a(4\,\mathrm{PRF}_{ch}/T_{sub})$; it writes the number in a block of $\pm h_w$ cells around each position — the target's and the copies' — and zero elsewhere. The kernel is the one of the look-up table: it does $f \to \beta \to x$ and reads a single cell per bin.



Neighbouring targets. Two targets closer than $2h_w$ have overlapping blocks, and the bins between them carry energy of both; there is no single right value there. The map merges them into one block, linearly interpolated between the targets (the natural choice: for continuous terrain ΔC_0 varies continuously with x). This is applied before the copies are drawn, so the merged block is copied as a whole.





Left: spectrum of the triangular window of $T_{sub} = 20$ samples; $99.71\,\%$ of the energy inside the 1st null, $99.956\,\%$ inside the 2nd (the block), $99.987\,\%$ inside the 3rd. Right: the ruler $a(f)$. In this scene $h_w = 82$ cells (100 Hz), $X = 405$ cells (6.23 km).

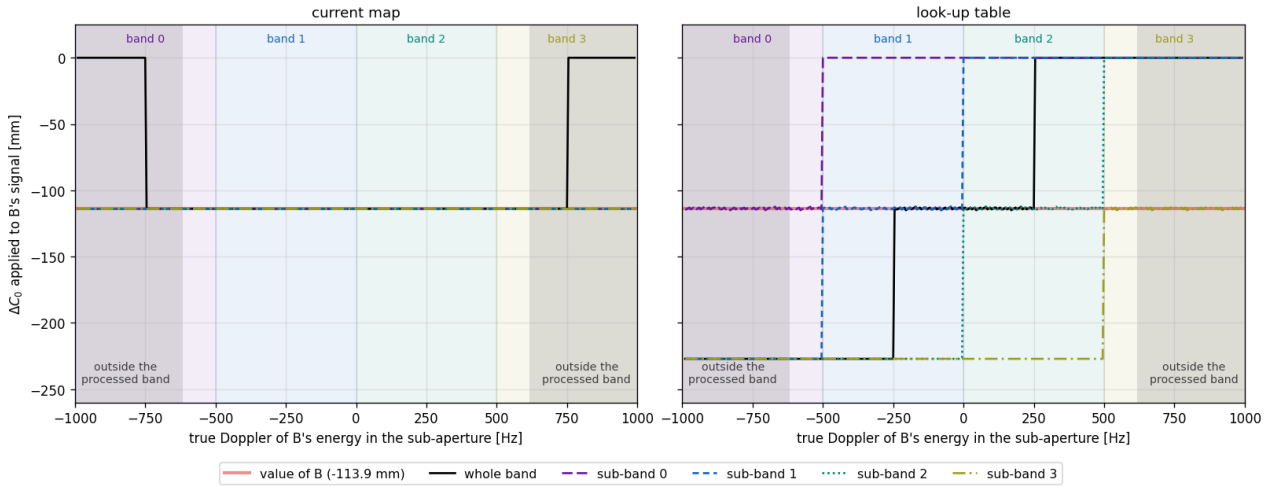
pipeline

targets $\rightarrow$ $\Delta C_0(\mathbf{p}_j)$ at the targets only $\rightarrow$ block $\pm h_w$ at $n_j + mX$, $m \in \mathcal{M}$ (nearest core wins where images overlap) $\rightarrow$ $\boxed{f \rightarrow \beta \rightarrow x}$ $\rightarrow$ $e^{+j2\pi M[x]/\lambda}$

3) The difference, and what it does

in one sentence

The difference between the two methods lies only in what is written at the fold cells: the look-up table writes there the terrain that is there; the current map writes there the number of the target that folded.



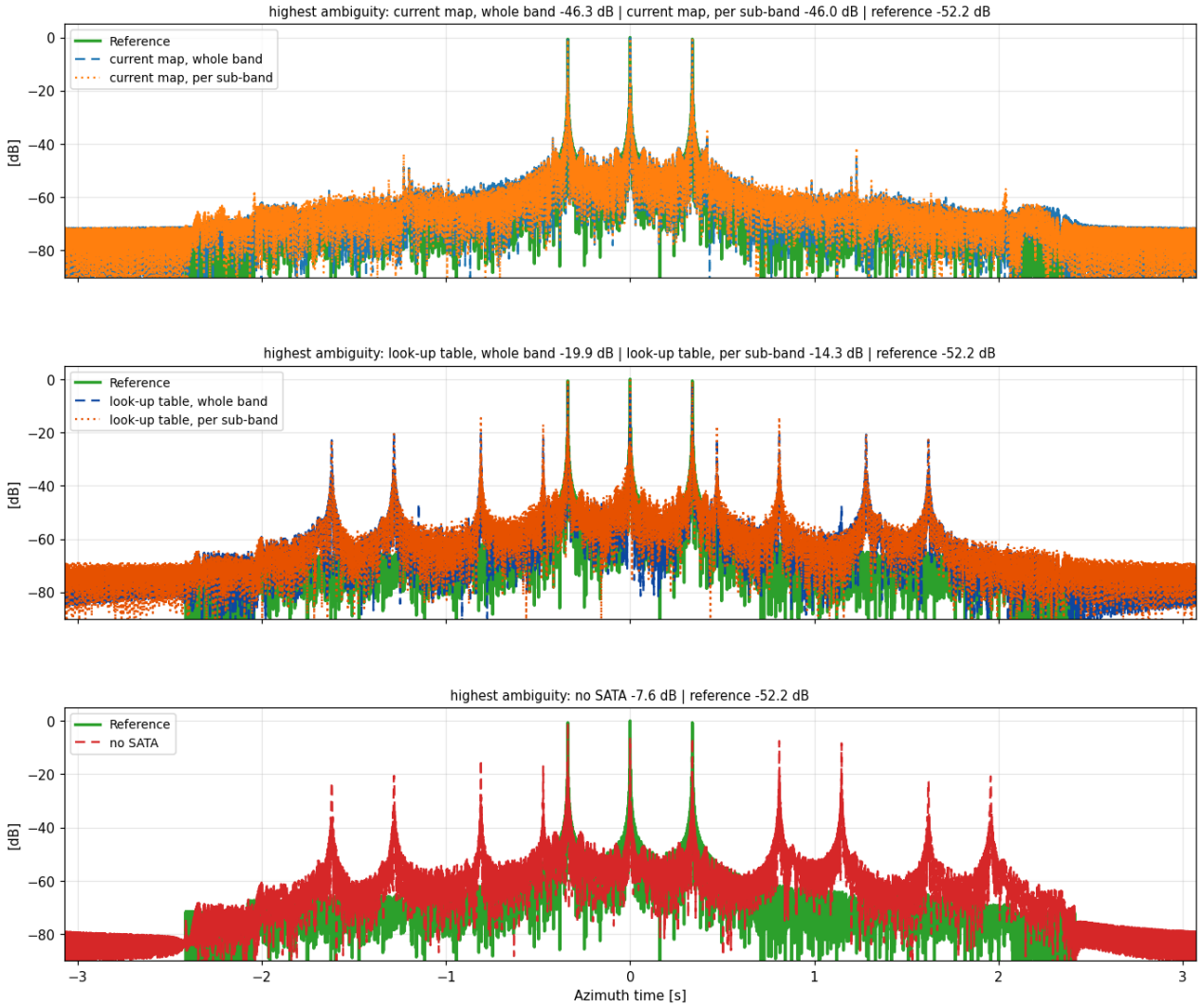
ΔC_0 applied to the energy of target B as a function of its true Doppler, for the whole-band kernel and the four sub-band kernels. Left: current map — every kernel applies B's value over the whole processed band, so whole band and sub-band give the same result. Right: look-up table — the whole-band kernel is right only for $|f| < \mathrm{PRF}_{ch}/2$ and reads the terrain $\pm X$ away elsewhere; sub-band k is right only inside band k.

highest ambiguity [dB]	1 target	A, B, C (2.6 km apart)	ramp, 5 targets 2.6 km	ramp, 5 targets 200 m
current map, whole / sub-band	-59.7 / -59.7	-46.3 / -46.0	-43.5 / -43.6	-36.5 / -36.5
look-up table, whole / sub-band	-63.6 / -63.6	-19.9 / -14.3	-23.2 / -18.0	-20.0 / -15.0
oracle (perfect phase)	-63.6	-45.9	-43.0	-50.9

- **Current map:** whole band = sub-band always, and equal to the oracle for isolated point targets. It relies on the fold cells being free (point targets, not $\approx X$ apart, no continuous terrain).

- **Look-up table:** exact when the terrain is the same $\pm X$ away (one target, flat terrain); otherwise each kernel corrects only part of the spectrum, the uncancelled parcels leak, and sub-band is worse than whole band because kernel k fixes only the parcel that goes to its own band.
- Neither solves the general case: a bin of the subsampled channel holds up to three ground positions and receives one phase.

4) IRFs, scene A, B, C



Azimuth compression with the matched filter of the reference point; 0 dB = peak of the Reference (ideal monostatic signal of the same targets, green). Top: current map, whole band and per sub-band — the two curves coincide. Middle: look-up table, whole band and per sub-band — the ambiguities at $t_j \pm 0.81$ s and ± 1.62 s come back, and sub-band is worse. Bottom: no SATA. Titles give the highest ambiguity of each curve.